XRPL Validator — Daily Engineering Report · 2026-08-22

Period: 2026-08-22 00:00 to 24:00 US Eastern (2026-08-22 04:00 to 2026-08-23 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): four fill-precision fixes — six consecutive ledgers byte-for-byte** (`b7831e7`, xrpl-validator) The full-ledger replay now reproduces #106455035-40 exactly (1,019 txs, root == account_hash). Four rules, each ~1 ulp or one nibble, all invisible to key-level gates: - payment: the post-round spend-side fee settlement is REMOVED — every walk fill already debits the sender the GROSS (CLOB move_leg_gross, AMM consume/consume_fib in_gross_rate, DirectHop rates), so the settlement charged the transfer rate a SECOND …
- **fix(xrpl-ledger): single-path re-arbitration in the bridge controller — seven consecutive ledgers** (`436102d`, xrpl-validator) The replay now reproduces #106455035-41 byte-for-byte (1,199 txs). Three rules from #106455041-42's maker-bot cluster: - offer: the bridged controller's direct-pair AMM turn now honors the per-round strand arbitration rippled does every driver iteration (activateNext drops strands whose upper bound misses limitQuality, then setMultiPath(activeStrands.size() > 1) — StrandFlow.h:672-674). A SINGLE-path round stops …
- **fix(xrpl-ledger): fresh multi-path verdict before the AMM turn — nine consecutive ledgers** (`f37bf9d`, xrpl-validator) The replay now reproduces #106455035-43 byte-for-byte (1,648 txs).
- **fix(xrpl-ledger): mulRatio requirements + blind-measurement guards — seventeen consecutive ledgers** (`fb798b3`, xrpl-validator) The replay now reproduces #106455035-51 byte-for-byte (~2,750 txs), and the mulRatio campaign lands whole: histsweep 426/426 INCLUDING the l106267220 specimen the first attempt regressed.
- **fix(xrpl-ledger): tec rolls back the AccountTxnID stamp; delivery trim rounds nearest — forty consecutive ledgers** (`36feb4f`, xrpl-validator) #106455035-74 now replay byte-for-byte (~5,600 txs).
- **fix(xrpl-ledger): settle the taker once per quality level, not per fill — forty-four consecutive ledgers** (`84e17e0`, xrpl-validator) rippled's flow settles the TAKER once per iteration: the fwd pass's DirectStep debits result.in — accumulated with per-offer STAmount adds — and only the MAKERS are settled per offer (consumeOffer's two sends, BookStep.cpp:900-915). An iteration is ONE quality level (or one AMM slice

standing in at a level head): BookStep stops at the level edge so the pass's marginal quality is well-defined, and flow() re-enters. …

- **fix(xrpl-ledger): a book redeems into the run behind it — forty-six consecutive ledgers** (`090b551`, xrpl-validator) A mixed strand's run saw only its own hops when deciding whether hop 0's issuer charges its transfer rate: qualitiesSrcIssues charges srcQOut exactly when the PREVIOUS STEP redeems, and hop 0 assumed strand-head (prev defaults to Issues). But a BOOK before the run reports DebtDirection::Redeems whenever ownerPaysTransferFee is off (BookStep.cpp:146), and OwnerPaysFee is a dormant amendment — on mainnet a book ALWAYS …
- **fix(xrpl-ledger): fwd hops reuse rev-stamped debt directions — fifty consecutive ledgers** (`84a026e`, xrpl-validator) rippled's DirectStepI::debtDirection returns the CACHE in the forward direction (DirectStep.cpp:492-499) — the direction stamped during the rev pass, when no funds had moved. Our run_fwd re-probed src_redeems from the live sandbox, so a hop that DRAINED its line flipped the next hop's view of it from Redeems to Issues mid-pass, and the redeem-charge vanished.

Independence program

No independence record for UTC day 2026-08-22; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Aggregated Oracle Feed for Vault Liquidations** (lending) Add a new OracleAggregateSet transaction and ltORACLE_AGGREGATE ledger object (keylet::oracleAggregate(owner, id)) holding a bounded STVector256 (max
- **Atomic Multilateral Netting Settlement** (settlement) Add a ttNETTING_SETTLE transaction carrying sfNettingCycle, a bounded STArray (max 20) of {Account, sfCheckID, sfAmount} entries describing a closed o
- **Attested Generic Messages over XChainBridge** (interop) Extend XChainAddClaimAttestation with optional sfMessageType and sfPayload (bounded Blob, max 1KB) SFields carried alongside the existing value-claim

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-22 04:26	—	xrpl-validator	`b7831e7`	fix(xrpl-ledger): four fill-precision fixes — six consecutive ledgers byte-for-byte
08-22 14:57	—	xrpl-validator	`436102d`	fix(xrpl-ledger): single-path re-arbitration in the bridge controller — seven consecutive ledgers
08-22 16:11	—	xrpl-validator	`f37bf9d`	fix(xrpl-ledger): fresh multi-path verdict before the AMM turn — nine consecutive ledgers
08-22 20:07	—	xrpl-validator	`fb798b3`	fix(xrpl-ledger): mulRatio requirements + blind-measurement guards — seventeen consecutive ledgers
08-22 21:32	—	xrpl-validator	`36feb4f`	fix(xrpl-ledger): tec rolls back the AccountTxnID stamp; delivery trim rounds nearest — forty consecutive ledgers
08-22 23:46	—	xrpl-validator	`84e17e0`	fix(xrpl-ledger): settle the taker once per quality level, not per fill — forty-four consecutive ledgers
08-23 01:16	—	xrpl-validator	`090b551`	fix(xrpl-ledger): a book redeems into the run behind it — forty-six consecutive ledgers
08-23 02:13	—	xrpl-validator	`84a026e`	fix(xrpl-ledger): fwd hops reuse rev-stamped debt directions — fifty consecutive ledgers

Times in UTC. A commit appears on the day it was committed to main or merged into it.